

Operation Blue Sentinel

Incident Response Report

Project Title

Operation Blue Sentinel: Building and Defending a Mini Security Operations Centre (SOC) Using Open-Source AI Security Tools

Team Name: Team A

Team Members and Roles

Team Member	Role
Olalekan	Security Engineer
Temidayo Salami	Incident Commander
John	SOC Analyst

Organised By:

The Purpose Protocol - Cohort 1

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1.0 Executive Summary

1.1 Project Overview

Cyber threats continue to increase as organizations rely more on digital systems and cloud services. To protect these systems, Security Operations Centers (SOCs) are used to continuously monitor networks, detect suspicious activities, and respond to security incidents before they become major security breaches.

Operation Blue Sentinel was designed to provide practical experience in building and operating a mini SOC using free and open-source cybersecurity tools hosted on cloud platforms. Throughout the project, the team built the environment, simulated a phishing attack, monitored security events, investigated alerts, and followed a structured incident response process based on industry best practices.

1.2 Scope of the Project

The project focused on building a cloud-based Security Operations Center (SOC) capable of monitoring security events and responding to simulated cyber threats. The environment included a Wazuh server hosted on **Microsoft Azure** and a phishing simulation server running **GoPhish** on a **Windows Server hosted in Oracle Cloud Infrastructure (OCI)**.

The project covered infrastructure deployment, security monitoring, threat detection, phishing simulation, incident response, documentation, and reporting. All activities were carried out in a controlled laboratory environment for educational purposes, and no production systems were involved.

1.3 Tools and Technologies Used

Tool / Technology	Purpose
Microsoft Azure	Hosted the Ubuntu virtual machine running the Wazuh server.
Oracle Cloud Infrastructure (OCI)	Hosted the Windows Server used for the GoPhish phishing simulation.
Ubuntu Server	Operating system for the Wazuh server.
Windows Server	Operating system hosting the GoPhish application.
Wazuh	Security Information and Event Management (SIEM) platform used for log collection, monitoring, and threat detection.

Wazuh Agent	Collected logs from monitored endpoints and forwarded them to the Wazuh server.
VirusTotal API	Performed file hash and IP reputation checks to enrich security alerts with threat intelligence.
GoPhish	Simulated phishing emails to test the team's ability to detect and respond to phishing attacks.
MITRE ATT&CK Framework	Used to understand and map attacker techniques during the investigation.
NIST SP 800-61 Incident Response Framework	Guided the incident response process from detection to recovery.
GitHub	Used for project documentation, collaboration, and version control.

2.0 Team Roles

The project was completed by a Team of three members. Each member was assigned a specific role based on the responsibilities defined in the project. Although each person had a primary responsibility, the team worked together throughout the project to ensure its successful completion.

Team Member	Role	Primary Responsibilities
Olalekan	Security Engineer	Set up and configured the SOC environment, installed and managed Wazuh, deployed the Wazuh agent, integrated security tools, and ensured the environment functioned correctly throughout the project.
Temidayo Salami	Incident Commander	Coordinated the project activities, documented the incident response process, maintained the incident log, made response decisions during the simulation, and prepared the final incident response report.
John	SOC Analyst	Monitored the Wazuh dashboard, investigated security alerts, analyzed logs and attack activities, mapped findings to the MITRE ATT&CK framework, and reported security observations.

2.1 Team Collaboration

Although each member had a defined role, the project was carried out collaboratively. The team held regular discussions, shared findings, verified configurations, and supported one another during system setup, attack simulation, incident analysis, and documentation. This collaborative approach helped ensure that tasks were completed accurately and that the project objectives were achieved successfully.

3.0 System Architecture

The project was built to simulate a simple Security Operations Center (SOC) for monitoring and responding to cyber threats. The environment included a **Wazuh server** for security monitoring, **two employee endpoints** for log collection, a **GoPhish server** for phishing simulation, and **VirusTotal** for threat intelligence.

The monitoring environment consisted of a Wazuh server hosted on Microsoft Azure, monitored Windows and Linux endpoints running the Wazuh agent, a GoPhish server hosted on Oracle Cloud Infrastructure for phishing simulation, and VirusTotal integration for file reputation checks. These components worked together to support centralized monitoring and incident detection.

The Wazuh server collected and analyzed logs from the endpoints, while GoPhish simulated phishing attacks. VirusTotal provided additional information about suspicious files and IP addresses detected during the investigation. **Figure 1** shows the overall architecture and how the different components communicate during the simulation.

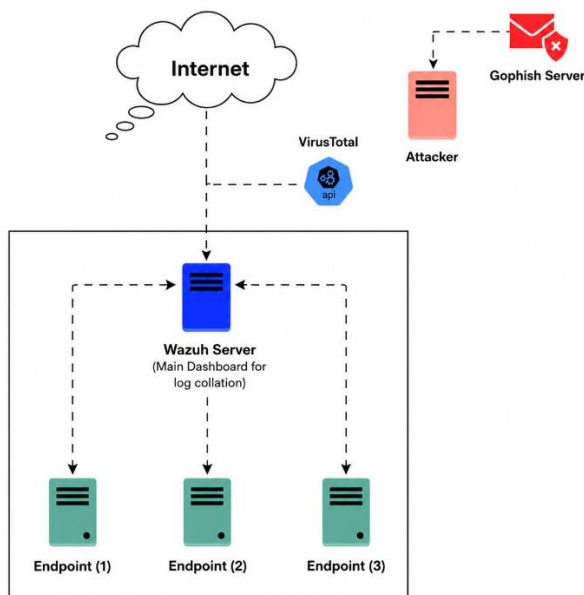


Figure 1: System Architecture of the Operation Blue Sentinel Environment.

4.0 Environment Setup

The project environment was deployed using cloud infrastructure and open-source cybersecurity tools. Each component was configured to perform a specific role in monitoring, detecting, and responding to security incidents.

4.1 Wazuh Server Deployment

The Wazuh server was deployed on a virtual machine hosted on **Microsoft Azure**. It served as the central Security Information and Event Management (SIEM) platform for the project.

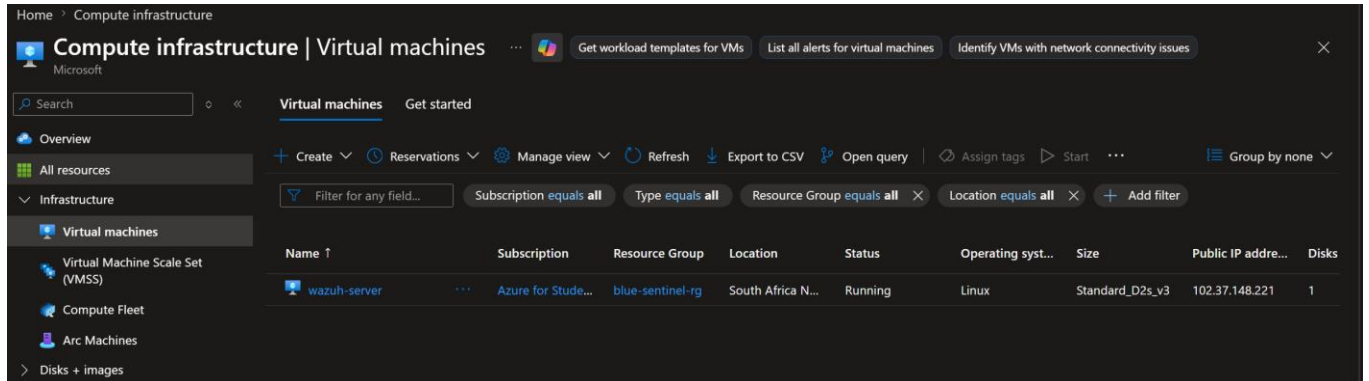


Figure 2: *Wazuh Sever running on Microsoft Azure*

The server was configured to:

- Collect security logs from monitored endpoints.
- Analyze incoming events.
- Generate security alerts.
- Display alerts through the Wazuh Dashboard.
- Integrate with VirusTotal for threat intelligence.

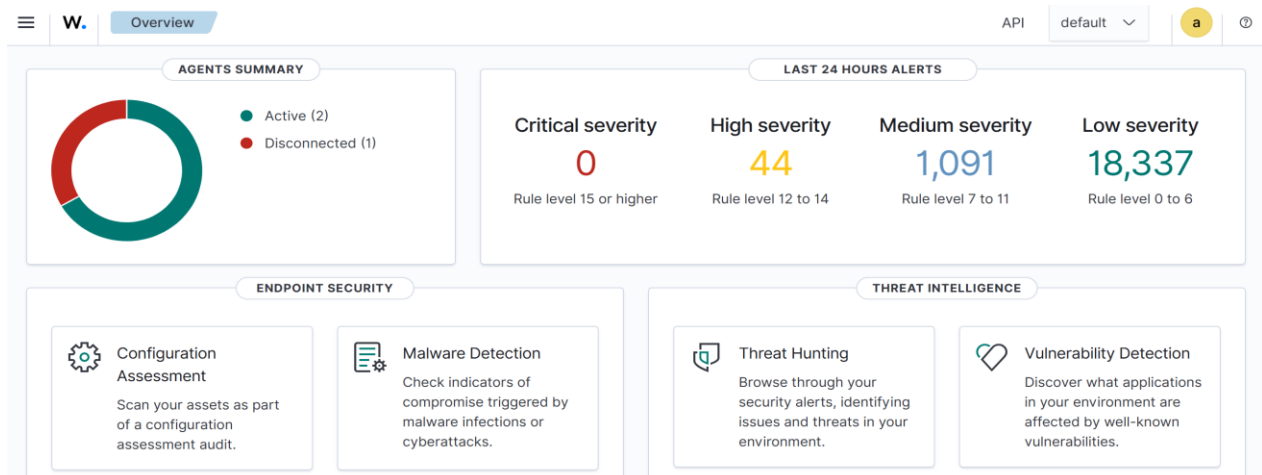


Figure 3: *Wazuh Dashboard showing Agent summary.*

4.3 Log Collection Configuration

The agent configuration was modified to collect Windows Security, Application and System logs as well as Linux authentication and system logs. File Integrity Monitoring (FIM) was also enabled to monitor changes within the Downloads directory.

```
<!-- Windows Log Collection -->
<agent_config os="Windows">
<!-- This tells your laptop to monitor the downloads folder in real-time -->
<syscheck>
  <directories check_all="yes"
  realtime="yes">C:\Users\Public\Downloads</directories>
</syscheck>
<!-- ensure you change C:\Users\Public\Downloads to the correct path of
your folder -->
  <localfile>
    <log_format>eventchannel</log_format>
    <location>Security</location>
  </localfile>
  <localfile>
    <log_format>eventchannel</log_format>
    <location>System</location>
  </localfile>
  <localfile>
    <log_format>eventchannel</log_format>
    <location>Application</location>
  </localfile>
  <syscheck>
    <directories check_all="yes"
    realtime="yes">C:\Users\handy\Downloads</directories>
  </syscheck>
</agent_config>
<!-- Linux Log Collection -->
<agent_config os="Linux">
  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/auth.log</location>
  </localfile>
  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/syslog</location>
  </localfile>
</agent_config>
```

Figure 4: Agent configuration was modified to collect logs

4.3 Endpoint Configuration

Three employee endpoints were configured as monitored systems within the internal network. A Wazuh agent was installed on each endpoint to collect security events and forward them to the Wazuh server.

These endpoints generated logs related to user authentication, file activities, and system events, allowing the SOC team to monitor their security status in real time.

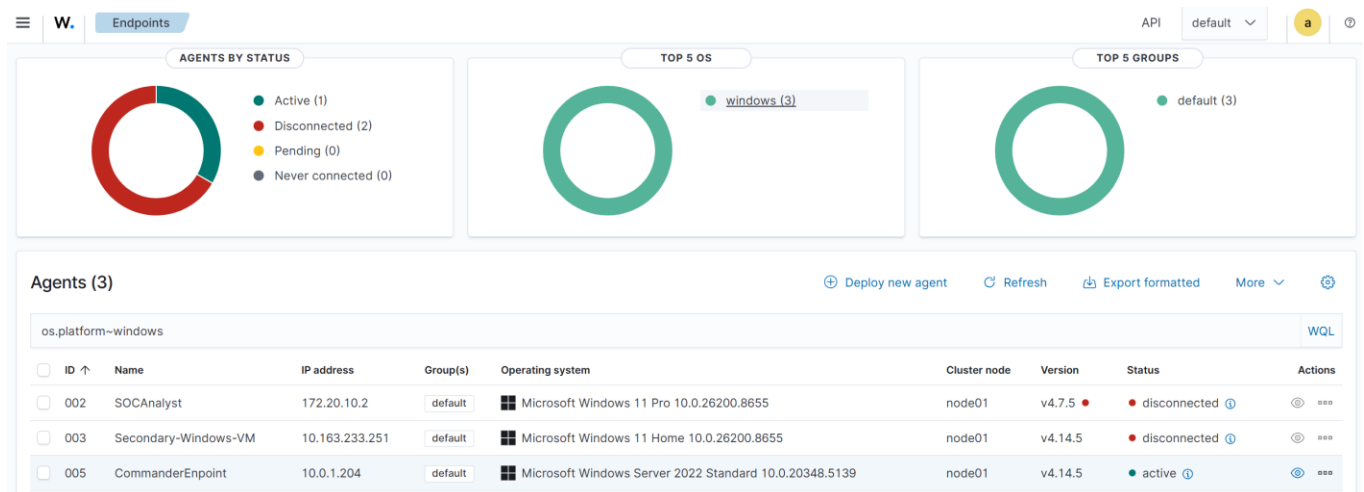


Figure 5: Connected Wazuh Agents.

4.4 GoPhish Server Setup

The GoPhish server was deployed on a **Windows Server hosted in Oracle Cloud Infrastructure (OCI)**. This server was used to simulate phishing attacks as part of the incident response exercise. The phishing emails created in GoPhish allowed the team to observe how phishing attacks can be delivered and how security monitoring tools respond to suspicious activities.

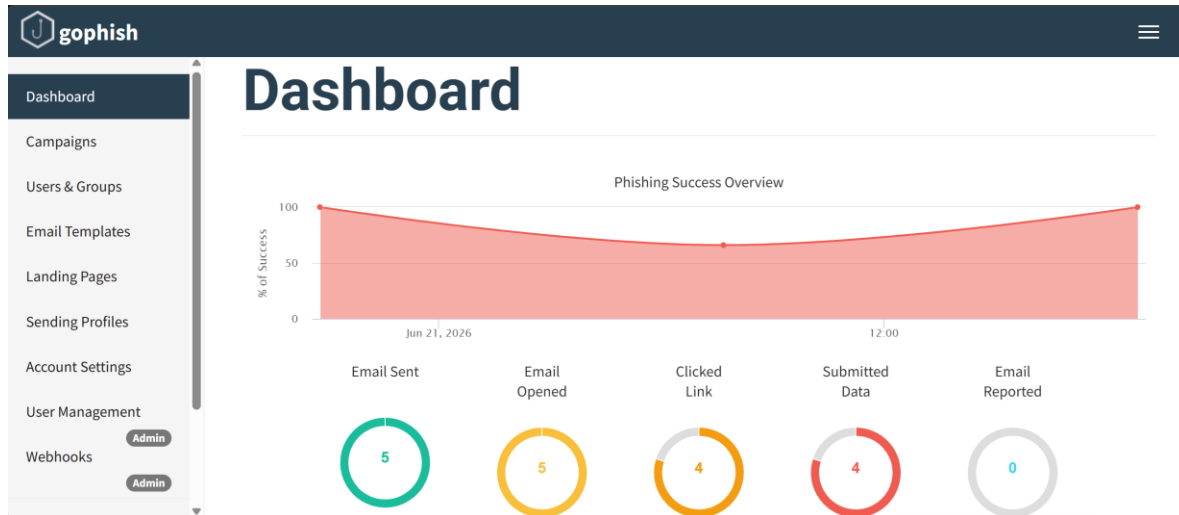


Figure 6: GoPhish Administration Dashboard.

4.4 VirusTotal Integration

The VirusTotal API was integrated with the Wazuh server to improve threat detection. During the investigation, Wazuh automatically checked the reputation of suspicious files and IP addresses against VirusTotal's threat intelligence database.

This integration provided additional information that helped the team determine whether detected activities were malicious or safe.

```
<integration>
  <name>virustotal</name>
  <api_key>YOUR_VIRUSTOTAL_API_KEY</api_key>
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>
```

Figure 7: VirusTotal Integration with Wazuh.

4.4 Active Response Configuration

Wazuh Active Response was configured to automatically execute a custom response whenever a malicious file was detected. A custom Windows script was created to remove suspicious files from the monitored Downloads directory after a matching alert was generated.



Figure 8: Active response configuration

4.5 Environment Verification

After completing the setup, the team verified that all components were communicating correctly.

The following checks were performed:

- The Wazuh server was running successfully.
- The monitored endpoints were connected and actively sending logs.
- Security events were visible on the Wazuh dashboard.
- VirusTotal integration was functioning correctly.
- The GoPhish server was accessible and ready for phishing simulation.

These verification steps confirmed that the environment was fully operational and ready for the attack simulation and incident response activities.

☐	ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
☐	002	SOCAnalyst	172.20.10.2	default	Microsoft Windows 11 Pro 10.0.26200.8655	node01	v4.7.5	disconnected	
☐	003	Secondary-Windows-VM	172.21.92.251	default	Microsoft Windows 11 Home 10.0.26200.8655	node01	v4.14.5	active	
☐	005	CommanderEndpoint	10.0.1.204	default	Microsoft Windows Server 2022 Standard 10.0.20348.5139	node01	v4.14.5	active	

Figure 9: Wazuh Dashboard showing connected agents

5.0 Attack Scenario

The objective of this project was to simulate a cyberattack and evaluate the team's ability to detect, investigate, and respond to security incidents using Wazuh.

The attack began with a phishing campaign created using **GoPhish**. Five phishing emails were sent to selected users within the simulated environment. Out of the five emails, four were opened, four users clicked the phishing link, and four users submitted their login credentials.

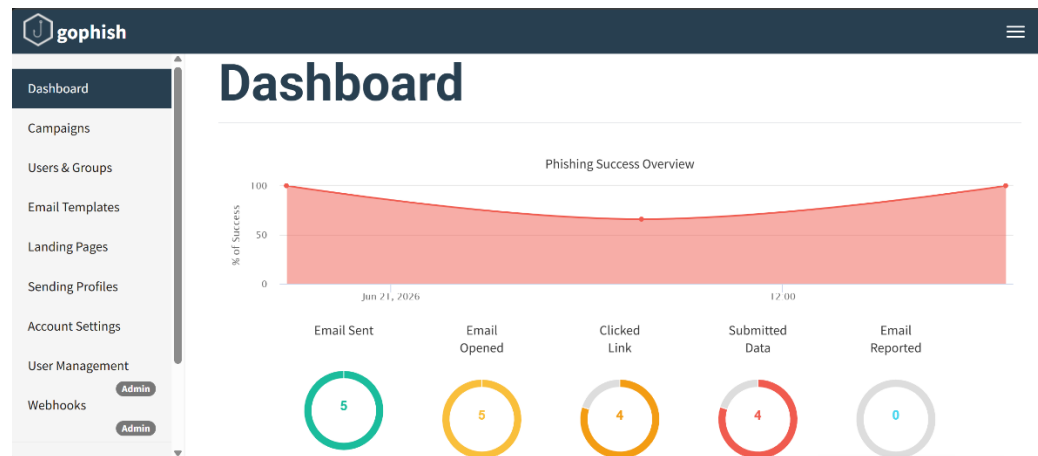


Figure 10: GoPhish Campaign Results.

After the phishing campaign, multiple unauthorized authentication attempts were detected on the Wazuh server. The attacker repeatedly attempted to log in to the server through SSH using different passwords, indicating a brute-force attack against the **root** account.

To support monitoring, Wazuh agents had already been deployed on the Windows endpoints. These agents collected authentication logs, Windows Event Logs, and File Integrity Monitoring (FIM) events. VirusTotal integration was also configured to verify the reputation of suspicious files detected during monitoring.

The collected logs were forwarded to the Wazuh Manager, where they were analyzed and converted into security alerts.

6.0 Incident Detection

The incident was detected through continuous monitoring using the Wazuh Security Information and Event Management (SIEM) platform. Wazuh collected logs from all monitored endpoints and generated alerts whenever suspicious activities were identified.

The first indication of suspicious activity was a high number of failed authentication attempts against the Wazuh server. These repeated login failures suggested a brute-force attack targeting the root account. During the same period, successful authentication events were also recorded.

The dashboard displayed:

- Total security alerts generated.
- Failed authentication attempts.
- Successful authentication attempts.

- Top attacking IP addresses.
- Alert severity.
- Authentication activity over time.

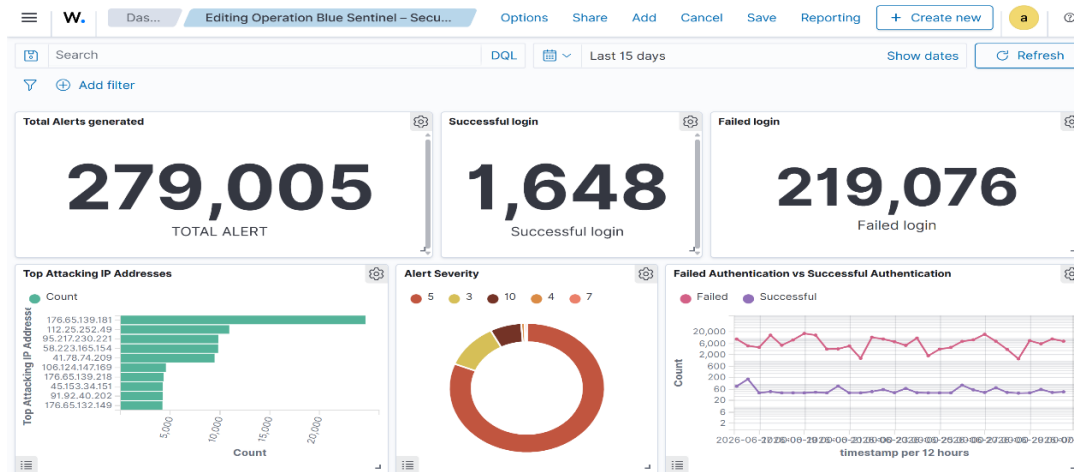


Figure 11: Operation Blue Sentinel Security Dashboard.

6.1 Threat Hunting

The SOC Analyst then performed a detailed investigation using the Wazuh Threat Hunting Dashboard. The analysis showed a very high number of authentication failures compared to successful logins, confirming an active brute-force attack against the server. The dashboard also showed attack spikes during specific periods, helping the team identify when the attack was most active.

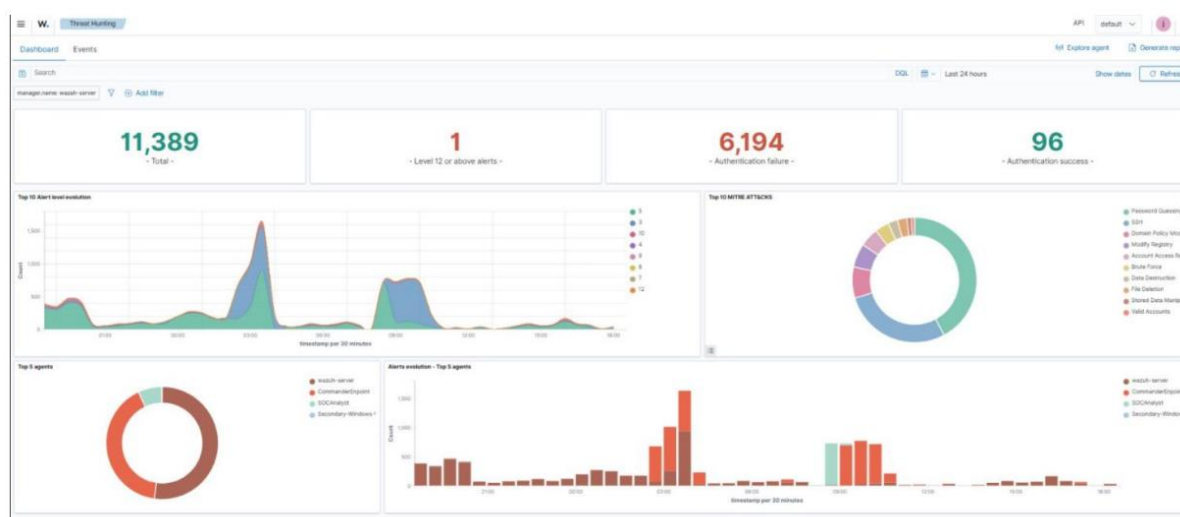


Figure 12: Threat Hunting Dashboard.

6.2 MITRE ATT&CK

The MITRE ATT&CK dashboard showed that **Defense Evasion** was the most common tactic detected during the monitoring period, indicating that the attacker attempted to avoid detection while carrying out malicious activities.

Among the attack techniques, **Modify Registry** generated the highest number of alerts, followed by **Stored Data Manipulation** and **File Deletion**. Other techniques, including **Data Destruction** and **Domain Policy Modification**, were detected less frequently.

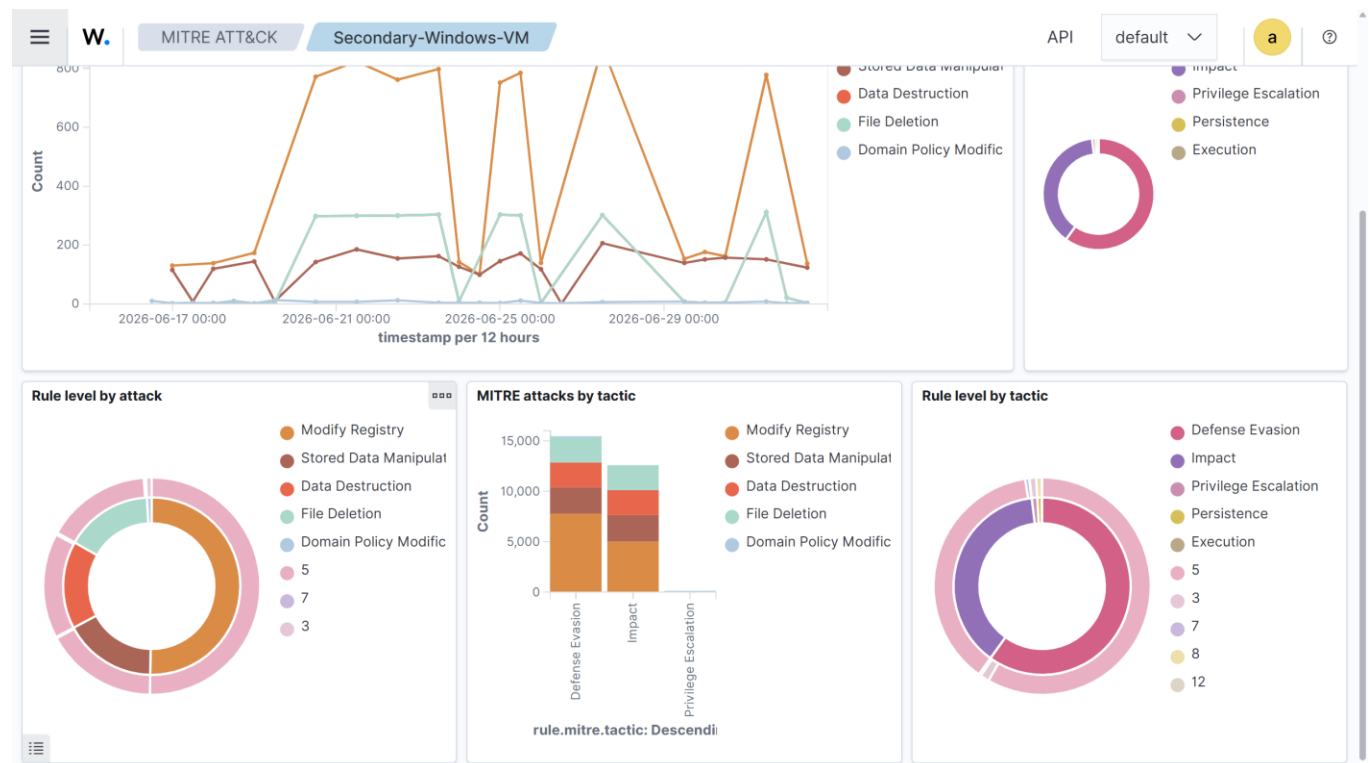


Figure 13: MITRE ATT&CK Dashboard.

6.3 Vulnerability Detection

The SOC Analyst also reviewed the Vulnerability Detection Dashboard. The analysis identified **146 vulnerabilities** across the monitored Windows endpoints, including **7 Critical**, **100 High**, and **33 Medium** Severity vulnerabilities. One Windows Server endpoint contained the highest number of vulnerabilities, making it the highest-risk asset within the environment.

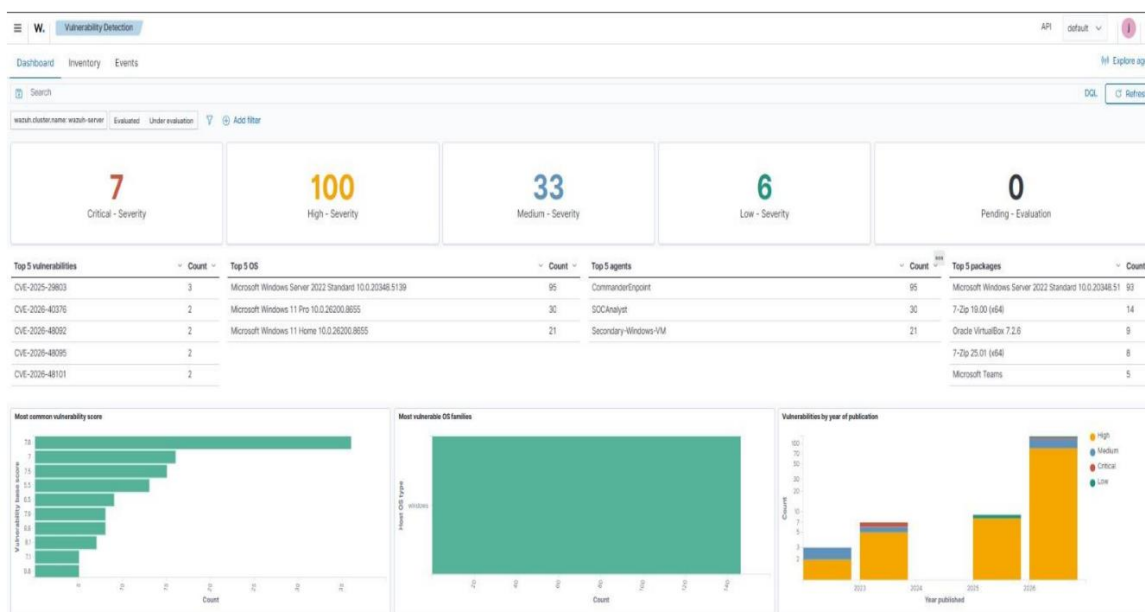


Figure 14: Vulnerability Detection Dashboard.

In addition to authentication monitoring, the configured Wazuh agents continuously collected Windows Event Logs, Linux authentication logs, and File Integrity Monitoring (FIM) events. VirusTotal integration was enabled to verify the reputation of suspicious files detected through File Integrity Monitoring, providing additional threat intelligence during the investigation.

Based on the collected evidence, the incident was classified as a High Severity security incident because it combined phishing activity, brute-force authentication attempts, successful logins, privilege escalation, and multiple critical vulnerabilities.

7.0 Incident Response

The investigation began by reviewing the authentication logs collected by Wazuh. Failed and successful login events were analyzed to identify the targeted accounts, source IP addresses, and the sequence of activities performed by the attacker.

The custom dashboard helped the Incident Commander monitor the overall security situation while the SOC Analyst performed detailed log analysis using the Threat Hunting and MITRE ATT&CK dashboards. Together, these tools provided a clear understanding of the attack progression and supported informed decision-making.

7.1 Authentication Logs

During the investigation, the engineer verified that all monitored endpoints were successfully reporting logs to the Wazuh Manager. The configured log collection settings ensured that Windows Security logs, Linux authentication logs, and File Integrity Monitoring events were available throughout the investigation.

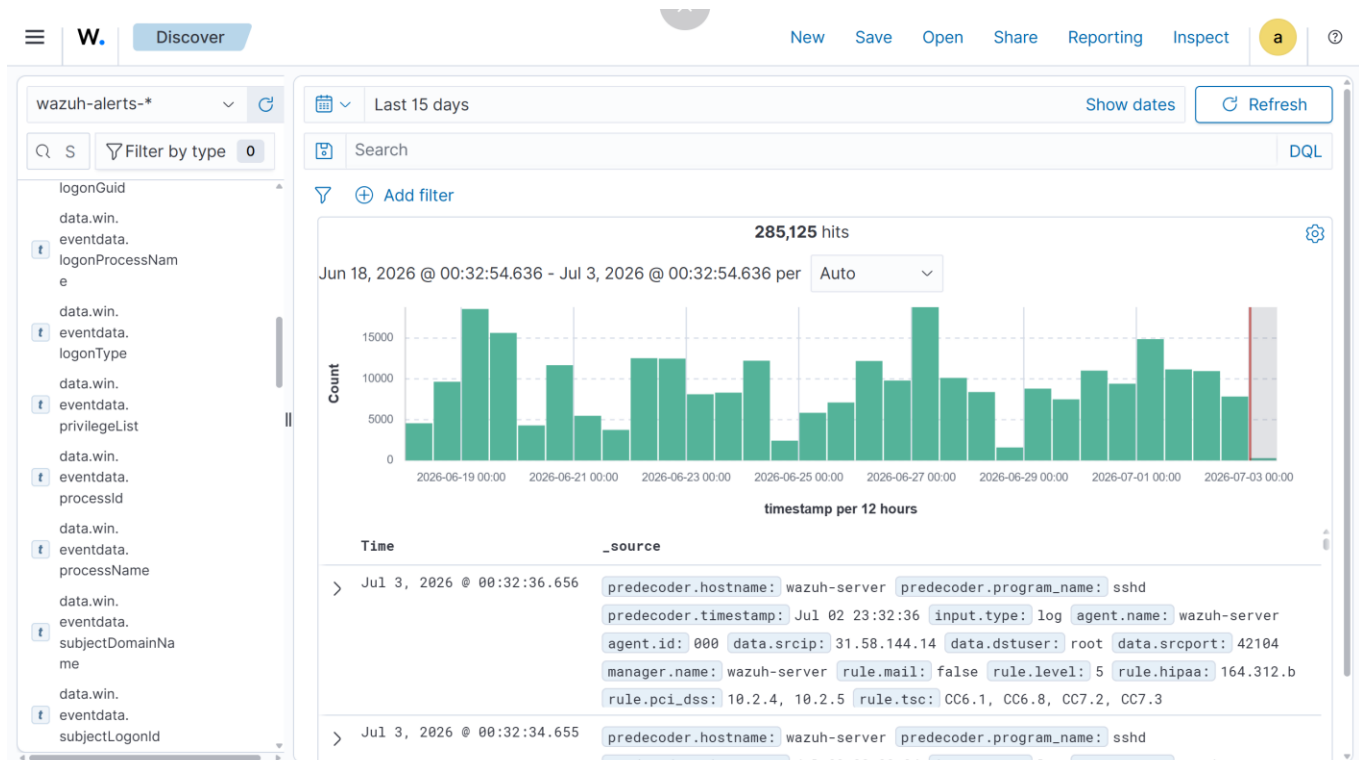


Figure 15: Authentication Events and Security Alerts.

The SOC Analyst recommended the following actions after completing the investigation:

- Review all successful authentication events.
- Investigate the detected privilege escalation activity.
- Block suspicious source IP addresses.
- Patch all Critical and High vulnerabilities.
- Perform File Integrity Monitoring checks for persistence.
- Continue monitoring the environment for additional suspicious activities.

The Incident Commander documented all findings, screenshots, and evidence throughout the investigation to support the final incident report and maintain a complete record of the incident response process.

8.0 Timeline

Time/Stage	Activity	Outcome
Environment Setup	Wazuh, GoPhish, agents, VirusTotal, and Active Response were configured.	Monitoring environment became operational.
Phishing Campaign	Five phishing emails were sent to users.	Four users opened the email, clicked the link, and submitted credentials.

Attack Execution	Multiple SSH login attempts targeted the Wazuh server.	Authentication failure alerts were generated.
Detection	Wazuh detected suspicious activities and generated security alerts.	SOC team began investigation.
Investigation	Dashboards, MITRE ATT&CK, Threat Hunting, and Vulnerability Detection were reviewed.	Attack techniques and affected systems were identified.
Response	Logs were analyzed, malicious IPs identified, and response actions recommended.	Incident documented and mitigation measures proposed.

9.0 Challenges

During the implementation of this project, the team encountered several technical challenges while setting up the environment, configuring the security tools, and monitoring the simulated attack. The major challenges are discussed below:

9.1 Cloud Environment Setup

One of the first challenges was setting up the cloud infrastructure. The project required multiple cloud servers, including Microsoft Azure for the Wazuh server and Oracle Cloud Infrastructure for the GoPhish server. Obtaining free-tier resources, configuring virtual machines, assigning public IP addresses, and ensuring connectivity between the servers required additional time before deployment could begin.

Despite these challenges, both cloud environments were successfully configured and used throughout the project.

9.2 Building the Wazuh Dashboard

Developing a dashboard that clearly represented the incident proved to be more difficult than expected. Wazuh stores security data across many fields, making it challenging to identify which fields were most suitable for visualization. Several visualizations initially produced incomplete or misleading results because of incorrect field selection and aggregation methods.

The team spent time exploring the available data and refining the dashboard until it accurately displayed authentication activity, alert severity, attack sources, and other key security events. This resulted in a dashboard that provided a clear summary of the incident and supported the investigation.

9.3 GoPhish Tracking

The phishing campaign was successfully delivered, and emails reached the intended recipients. However, although users clicked the phishing links, some interactions were not immediately reflected on the GoPhish dashboard. This made it difficult to accurately monitor user activity during the simulation.

The issue was investigated by reviewing the GoPhish configuration and network settings. After making the necessary adjustments, the campaign tracking functioned as expected and the required campaign statistics were successfully captured.

9.4 Wazuh Configuration and Integration

Configuring Wazuh required several manual adjustments before all components worked correctly. Log collection, agent deployment, VirusTotal integration, File Integrity Monitoring (FIM), and Active Response all required additional configuration beyond the default installation.

Differences between Wazuh software versions also meant that some deployment steps differed from the available documentation, requiring additional testing before the desired results were achieved.

9.5 Incident Analysis

During the investigation, the large number of authentication alerts made it difficult to quickly distinguish normal activity from malicious activity. The SOC Analyst had to correlate information from multiple dashboards, including Threat Hunting, MITRE ATT&CK, and Vulnerability Detection, before confirming the attack pattern.

This highlighted the importance of combining different sources of security information rather than relying on a single dashboard during incident investigations.

9.6 Team Coordination

Since the project involved different responsibilities, including infrastructure deployment, security monitoring, phishing simulation, and incident management, effective communication among team members was essential. Regular updates and collaboration ensured that findings from the engineer, SOC Analyst, and Incident Commander were combined into a single incident response process and final report.

10.0 Results

The project successfully demonstrated the deployment and use of a Security Information and Event Management (SIEM) solution for detecting and responding to cyber threats. The Wazuh environment was successfully deployed, monitored endpoints were connected, and logs were collected from both Windows and Linux servers.

The phishing campaign carried out using GoPhish showed that four out of five users interacted with the phishing email and submitted their credentials.

Wazuh successfully detected multiple failed SSH authentication attempts, mapped the activities to the MITRE ATT&CK framework, identified vulnerabilities on monitored endpoints, and generated alerts that supported the investigation. The custom dashboard also provided a clear overview of the incident.

11.0 Lessons Learned

This project demonstrated the importance of continuous security monitoring in detecting cyber threats before they cause significant damage. It also showed that collecting logs alone is not enough; the ability to analyze and present security information through meaningful dashboards greatly improves incident investigation.

The project also reinforced the importance of proper system configuration, regular vulnerability management, and teamwork during incident response. Collaboration between the Incident Commander, Security Engineer, and SOC Analyst enabled the team to successfully detect, investigate, and document the simulated attack.

12.0 Recommendations

Based on the findings from this project, the following recommendations are proposed:

- Regularly apply security patches to reduce system vulnerabilities.
- Enable continuous monitoring using SIEM solutions such as Wazuh.
- Conduct regular phishing awareness training for users.
- Strengthen authentication by implementing Multi-Factor Authentication (MFA).
- Restrict unnecessary public access to critical servers.
- Continuously review security dashboards and investigate high-severity alerts promptly.
- Regularly update threat intelligence integrations such as VirusTotal.

13.0 Conclusion

This project successfully demonstrated the complete incident response lifecycle, from environment setup and attack simulation to detection, investigation, and response. By combining Wazuh,

GoPhish, VirusTotal, and custom dashboards, the team was able to monitor security events, identify malicious activities, and document appropriate response actions.

Overall, the project provided practical experience in security monitoring, incident handling, and teamwork while demonstrating how modern security tools can improve an organization's ability to detect and respond to cyber threats.

14.0 References

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